

IVAN GUNAWAN SIMANJUNTAK

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Jl.Sibolahotang SAS, Balige, Toba, Sumatera Utara

Final-year Computer Technology student at Institut Teknologi Del with a strong academic and practical foundation in software development, network security, and IoT systems.

I have hands-on experience leading and managing academic projects, including the development of web-based applications, IoT-based book location systems using ESP32 BLE, and cybersecurity research focused on anomaly-based DDoS detection using Wazuh SIEM.

Through these projects, I have developed solid skills in Linux environments, web development (HTML, CSS, JavaScript, PHP), and basic network infrastructure and security concepts.

I am accustomed to working with structured workflows, collaborating in team-based projects, and translating technical requirements into functional solutions.

As a motivated final-year student seeking an internship opportunity, I am eager to apply my technical knowledge in a real-world professional environment while continuing to learn and grow.

I am particularly interested in roles related to software engineering, backend development, network infrastructure, or cybersecurity, where I can contribute effectively, gain industry experience, and deliver meaningful impact.

Project Experiences

Del Institute of Technology - Laguboti, Sumatera Utara, Indonesia

Jan 2024 - Jun 2024

Project Manager and Developer

Web-Based Application Development (Academic Projects) :

- Led the end-to-end development of a responsive web-based catalog application featuring pricing and search functionality.
- Designed and implemented a clean, user-friendly UI/UX to improve accessibility and usability.
- Coordinated team workflows, task distribution, and project timelines to ensure timely and structured delivery.
- Contributed to backend implementation and system logic using web development technologies.

Del Institute of Technology - Laguboti, Sumatera Utara, Indonesia

Jan 2025 - Jun 2025

Project Manager and Developer

BLE-Based Book Location System in Perpustakaan IT Del using ESP-32 :

- Developed an IoT-based book location system using ESP32 microcontrollers to detect and broadcast BLE signals.
- Implemented BLE scanning logic to identify nearby devices and determine book shelf locations.
- Integrated hardware systems with a mobile application interface to visualize real-time book locations.
- Ensured reliable communication between embedded devices and application layers through structured testing.

Del Institute of Technology - Laguboti, Sumatera Utara, Indonesia

Nov 2025 - Jan 2026

Research Lead and Developer

Real-time Automation System for DDoS Based on Anomaly Detection and Route Nullification :

- Led academic research on cybersecurity threats with a focus on anomaly-based DDoS detection.
- Designed and implemented a real-time monitoring system using Wazuh SIEM to identify abnormal traffic patterns.
- Developed automated route nullification mechanisms to mitigate detected attacks in real time.
- Analyzed security events and alerts to improve detection accuracy and system response effectiveness.

Del Institute of Technology - Laguboti, Sumatera Utara, Indonesia

Jan 2026 - Present

Full Stack & AI Engineer

Leading the development of "TaskPro", an enterprise-grade AI-powered task management system designed to enhance team collaboration and individual productivity.

- Engineered a scalable Full Stack application ("TaskPro") utilizing Next.js for a responsive frontend and Python FastAPI for a high-performance backend architecture.
- Integrated Generative AI (Google Gemini) to create a context-aware chatbot assistant that analyzes user tasks and provides real-time productivity strategies and prioritization.
- Implemented Real-Time Collaboration features using WebSockets, enabling instant synchronization of the Kanban board updates across multiple devices without page reloads.
- Designed a professional UI/UX with dual-theme support (Dark/Light Mode) and interactive data visualization charts to track weekly productivity metrics.

- Engineered an enterprise-ready productivity platform integrating Next.js (Frontend) and Python FastAPI (Backend) with a PostgreSQL database.
- Developed a "Context-Aware" AI Assistant using Google Gemini API, capable of analyzing active task data to provide personalized scheduling advice and motivation.
- Built a Real-Time Synchronization System using WebSockets, allowing seamless drag-and-drop interactions on the Kanban board for instant team collaboration.
- Implemented secure authentication utilizing OAuth2 with JWT (JSON Web Tokens) and bcrypt password hashing to ensure user data security.
- Designed comprehensive analytics dashboards using Recharts to visualize productivity trends, task completion rates, and weekly summaries.

Education

Del Institute of Technology - Laguboti, Sumatera Utara, Indonesia
Diploma in Associate Degree in Computer Technology

Aug 2023 - Sep 2026 (Expected)

Organizational Experience

BEM Del Institute of Technology - Laguboti, Sumatera Utara, Indonesia
Member

Aug 2023 - Present

Participated as a member of the Student Executive Board (BEM) in the Arts and Culture department.

- Contributed to the planning and execution of the Del Art Competition, managing participant registration.
- Contributed to talent scouting and program development for the Del Music Corner.
- Played a key role in designing and executing the arts exhibition segment of the Cultural Festival.
- Actively participated in student organization activities within the Arts and Culture division.

Skill

- **Hard Skills:** Programming | WordPress | Laravel | Database | Network Routing | Network Security | Penetration Testing | Embedded System | Internet of Things | Virtualization | Cloud Computing | Application Mobile
- **Soft Skills:** Critical Thinking | Problem Solving | Collaboration | Teamwork | Innovative | Communicative | Project Leadership & Management | Strategic Prioritization | Technical Communication